

materials including properties and uses of these.

Unit- 3.0**8 hrs**

Introduction to Material Testing covering, What is the “ Material Engineering” ?; Mechanical behavior and mechanical characteristics; Elasticity – principle and characteristics; Plastic deformation of metals; Tensile test – standards for different material (brittle, quasi-brittle, elastic and so on) True stress – strain interpretation of tensile test.

Unit- 4.0**9 hrs**

Hardness tests; Bending and torsion test; strength of ceramic; Internal friction, creep – fundamentals and characteristics; Brittle fracture of steel – temperature transition approach; Background of fracture mechanics; Discussion of fracture toughness testing – different materials; concept of fatigue of materials; Structural integrity assessment procedure and fracture mechanics.

Unit- 5.0**7 hrs**

Standard Testing & Evaluation Procedures covering, Laboratory for mechanical testing; Discussion about mechanical testing; Naming systems for various irons, steels and nonferrous metals; Discussion about elastic deformation; Plastic deformation; Impact test and transition temperatures.

Unit- 6.0**5 hrs**

Fracture mechanics – background; Fracture toughness – different materials; Fatigue of material; Creep.

Text/ Reference:-

1. Chudley, R., Greeno (2006), 'Building Construction Handbook' (6th ed.), R. Butterworth- Heinemann.
2. Khanna, S.K., Justo, C.E.G and Veeraragavan, A, ' Highway Materials and Pavement Testing', Nem Chand & Bros, Fifth Edition.
3. Various related updated & recent standards of BIS, IRC, ASTM, RILEM, AASHTO, etc. corresponding to materials used for Civil Engineering applications.
1. Kyriakos Komvopoulos (2011), Mechanical Testing of Engineering Materials, Cognella.
5. E.N. Dowling (1993), Mechanical Behaviour of Materials, Prentice Hall International Edition
2. American Society for Testing and Materials (ASTM), Annual Book of ASTM Standards (post 2000).
7. Related papers published in international journals.

Course Code- 119307**Universal Human Values****3 0 0 3****Unit- 1.0:****7 hrs****Introduction to Value Education:**

Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education), Understanding Value Education Sharing about Oneself, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Exploring Human Consciousness, Happiness and Prosperity – Current Scenario, Method to Fulfill the Basic Human Aspirations,

Unit- 2.0:**7 hrs****Harmony in the Human Being:**

Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self,